

MUST lecture 5

One- and two-factor Multivariate Analysis of Variation (MANOVA1 & MANOVA2)

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

One-factor MANOVA1 (g populations categorized by a single factor, indexed by l)

Model with g populations $X_{lj} \sim N_p(\mu_l, \Sigma_l), \quad l = 1, \dots, g, \quad j = 1, \dots, n_l, \quad n = \sum_{l=1}^g n_l$
 $\forall \Sigma_l = \Sigma \quad (\text{homoscedasticity}) \quad .$
 $\mu_l = \mu + \tau_l$

Hypothesis $H_0: \forall \tau_l = 0 \quad \text{against} \quad H_1: \exists \tau_l \neq 0$

Sample means overall $\bar{X} = \frac{1}{n} \sum_{l=1}^g \sum_{j=1}^{n_l} X_{lj}$
population-wise $\bar{X}_l = \frac{1}{n_l} \sum_{j=1}^{n_l} X_{lj}$

SS decomposition $SS_T = SS_W + SS_B$
 $SS_T = \sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X})(X_{lj} - \bar{X})^T \quad (\text{total})$
 $SS_W = \sum_{l=1}^g \sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l)(X_{lj} - \bar{X}_l)^T \quad (\text{within populations})$
 $SS_B = \sum_{l=1}^g \sum_{j=1}^{n_l} (\bar{X}_l - \bar{X})(\bar{X}_l - \bar{X})^T \quad (\text{between populations})$

Degrees of freedom SST: n-1 SSW: n-g SSB: g-1

Test statistics and test: see MANOVA1 test-statistics pdf-file

Accept of H_0

$$X_{lj} \sim N_p(\mu, \Sigma)$$

$$\widehat{\mu} = \bar{X}$$

$$\widehat{\Sigma} = S_T = \frac{SS_T}{n-1}$$

Rejected H_0

$$X_{lj} \sim N_p(\mu_l, \Sigma)$$

$$\widehat{\mu}_l = \bar{X}_l$$

$$\widehat{\Sigma} = S_W = \frac{SS_W}{n-g}$$

Pairwise Bonferroni CI's for $(\mu_{ki} - \mu_{li}), \quad k, l \in \{1, 2, \dots, g\}, \quad i \in \{1, 2, \dots, p\}$

Number of population pairs: $\binom{g}{2} = \frac{1}{2}g(g-1)$

Number of variables:

Total number of CI's: $\frac{1}{2}g(g-1)p$

CI for $(\mu_{ki} - \mu_{li})$: $\left[(\bar{X}_{ki} - \bar{X}_{li}) \pm t(n-g) \frac{\alpha}{g(g-1)p} \sqrt{\left(\frac{1}{n_k} + \frac{1}{n_l} \right) S_{W,ii}} \right], \quad k, l \in \{1, 2, \dots, g\}, \quad i \in \{1, 2, \dots, p\}$

Bartlett test for homoscedasticity

Model $X_{lj} \sim N_p(\mu_l, \Sigma_l), \quad l = 1, \dots, g, \quad j = 1, \dots, n_l, \quad n = \sum_{l=1}^g n_l$

Hypothesis $H_0: \forall \Sigma_l = \Sigma$ against $H_1: \exists \Sigma_{l_1} \neq \Sigma_{l_2}$

Test statistic $T = c \cdot \{(n - g) \log |S_P| - \sum_{l=1}^g (n_l - 1) \log |S_l|\}$

where the finite population correction factor

$$c = 1 - \frac{2p^2 + 3p - 1}{6(p+1)(g-1)} \left(\sum_{l=1}^g \frac{1}{n_l - 1} - \frac{1}{n - g} \right) \cong 1 \quad (\text{usually})$$

Approximately $T \sim \chi^2(\frac{1}{2}p(p+1)(g-1))$

Test reject $H_0 \Leftrightarrow T > \chi^2(\frac{1}{2}p(p+1)(g-1))_\alpha$

Two-factor MANOVA2 (g*b populations categorized by 2 factors, indexed by l and k)

Model with g*b populations

$$X_{lkj} \sim N_p(\mu_{lk}, \Sigma_{lk}), \quad l = 1, \dots, g, \quad k = 1, \dots, b, \quad j = 1, \dots, n$$
$$\forall \Sigma_{lk} = \Sigma \quad (\text{homoscedasticity}) \quad .$$
$$\mu_{lk} = \mu + \tau_l + \beta_k + \gamma_{lk}$$

Sample means

overall $\bar{X} = \frac{1}{gbn} \sum_{l=1}^g \sum_{k=1}^b \sum_{j=1}^n X_{lkj}$

population-wise $\bar{X}_{lk} = \frac{1}{n} \sum_{j=1}^n X_{lkj}$

column-wise $\bar{X}_{l\blacksquare} = \frac{1}{bn} \sum_{k=1}^b \sum_{j=1}^n X_{lkj} \quad (\text{factor 1})$

row-wise $\bar{X}_{\blacksquare k} = \frac{1}{gn} \sum_{l=1}^g \sum_{j=1}^n X_{lkj} \quad (\text{factor 2})$

SS decomposition

$$SS_T = SS_W + SS_{B1} + SS_{B2} + SS_{INT}$$
$$SS_T = \sum_{l=1}^g \sum_{k=1}^b \sum_{j=1}^n (X_{lkj} - \bar{X})(X_{lkj} - \bar{X})^T \quad (\text{total})$$
$$SS_W = \sum_{l=1}^g \sum_{k=1}^b \sum_{j=1}^n (X_{lkj} - \bar{X}_{lk})(X_{lkj} - \bar{X}_{lk})^T \quad (\text{within populations})$$
$$SS_{B1} = \sum_{l=1}^g \sum_{k=1}^b \sum_{j=1}^n (\bar{X}_{l\blacksquare} - \bar{X})(\bar{X}_{l\blacksquare} - \bar{X})^T \quad (\text{between columns})$$
$$SS_{B2} = \sum_{l=1}^g \sum_{k=1}^b \sum_{j=1}^n (\bar{X}_{\blacksquare k} - \bar{X})(\bar{X}_{\blacksquare k} - \bar{X})^T \quad (\text{between rows})$$
$$SS_{INT} = \sum_{l=1}^g \sum_{k=1}^b \sum_{j=1}^n (\bar{X}_{lk} - \bar{X}_{l\blacksquare} - \bar{X}_{\blacksquare k} + \bar{X})(\bar{X}_{lk} - \bar{X}_{l\blacksquare} - \bar{X}_{\blacksquare k} + \bar{X})^T \quad (\text{interaction})$$

Degrees of freedom

SST: $gbn-1$
SSW: $gb(n-1)$
SSB1: $g-1$
SSB2: $b-1$
SSINT: $(g-1)(b-1)$

Null hypotheses, test statistics and tests for

- no systematic interaction
- significant factor 1 effect
- significant factor 2 effect

See MANOVA2 test-statistics pdf-file

Bartlett test for MANOVA2 homoscedasticity straight forward generalization of the test for MANOVA1 homoscedasticity